

STATISTICS

Supplementary Document: Central Limit Theorem

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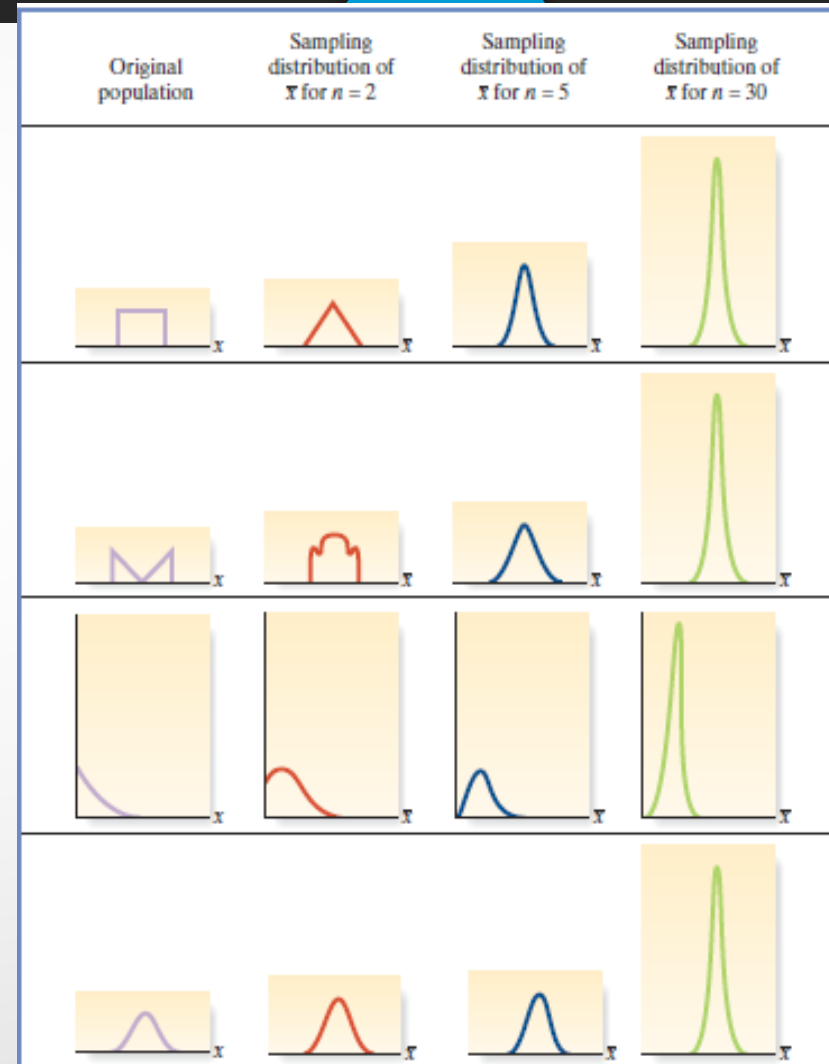
Rev:31.10.2023

Central Limit Theorem

Statistics

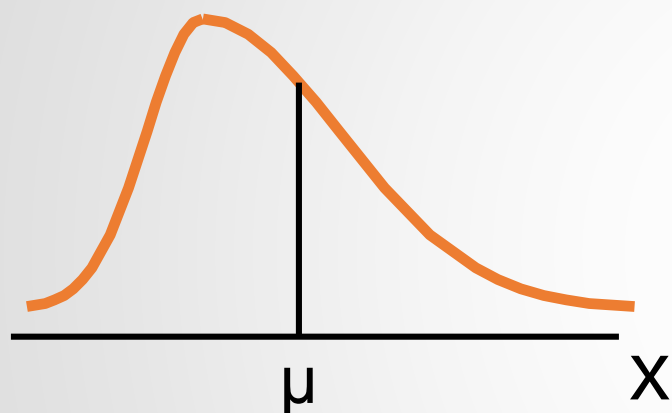
As sample size (n) increases, the sampling distribution of sample means ($\bar{X}$) approximates a normal distribution.

When a population has more skewness, a larger sample size is required to approximate a normal distribution.



Central Limit Theorem

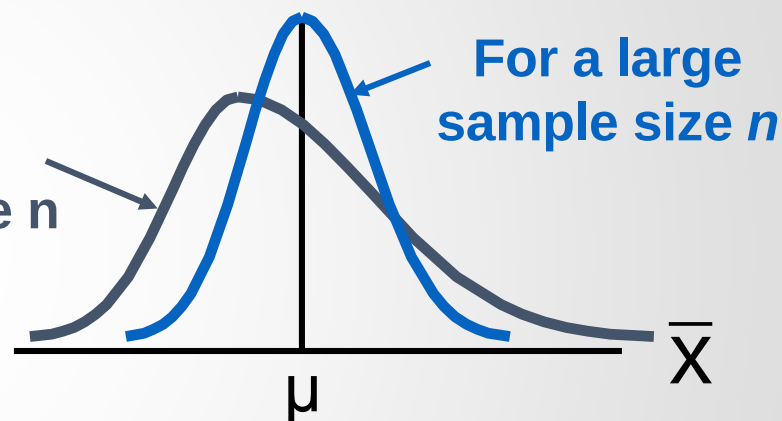
Statistics



$$E(X) = \mu$$

$$\sigma_X = \sigma$$

For a small
sample size n



$$E(\bar{X}) = \mu$$

$$\sigma_{\bar{X}} = \sigma / \sqrt{n}$$

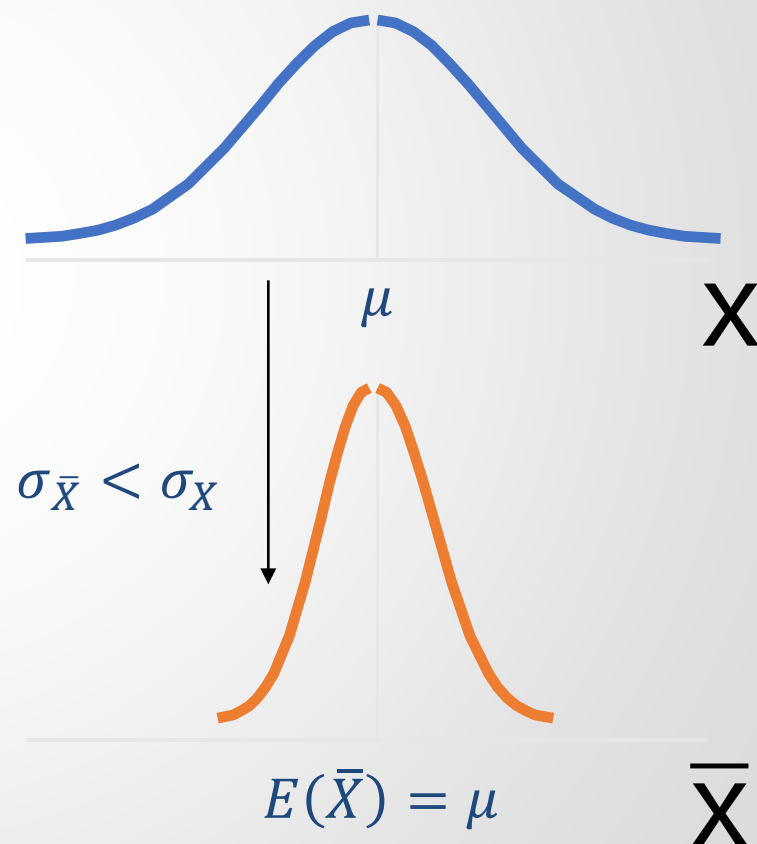
As sample size (n) increases, the sampling distribution of sample means ($\bar{X}$) approximates a normal distribution with mean μ and standard deviation $\sigma / \sqrt{n}$.

Central Limit Theorem

Statistics

Consider a population having a normal distribution with mean μ and standard deviation σ .

Since $\sigma_{\bar{X}} = \sigma / \sqrt{n}$,
as n increases, $\sigma_{\bar{X}}$ decreases.



Example

Statistics

Example: Let a coin is tossed and X be the number of occurring head. If the coin is tossed two times ($n=2$), show the sampling distribution of sample means. Then, compute $E(\bar{X})$ and $\sigma_{\bar{X}}$ by

- using sampling distribution
- probability theory

Tossing a coin 2 times ($n=2$)		
Sonuç (Atış 1, Atış 2)	Probability $f(\bar{X})$	$\bar{X}$
Head-Head ($x = 1, x = 1$)	$1/4$	1
Head-Tail ($x = 1, x = 0$)	$1/4$	0.5
Tail-Head ($x = 0, x = 1$)	$1/4$	0.5
Tail-Tail ($x = 0, x = 0$)	$1/4$	0